Exact Spectral Function of the Half-Filled Hubbard Dimer with a Site-Splitting Potential ΔV

gGut_mau project — `scan_ed_deltaV/`

October 19, 2026

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Purpose. This note generalizes the exact (ED) spectral function of the half-filled Hubbard dimer, derived in `/home/marm314/C_MEJUTO/gGut_notes/gGut/hubbard_dimer/gGut_notes/gGut.tex` (Sec. "Spectral function", Appendix "Exact spectral function of the dimer", its Eq. (eq:Aw_exact)) to the case where the two sites additionally carry a splitting potential ΔV (site 0 raised, site 1 lowered, by $\Delta V/2$). It contains ONLY exact diagonalization (ED) results: no Gutzwiller/gGA approximation enters anywhere in this note. Every closed-form formula below was derived symbolically (computer-algebra, not by hand) and then validated to machine precision ($< 10^{-14}$) against a fully independent, brute-force numerical exact diagonalization of the full $2^4 = 16$-dimensional Fock space (`exact_dimer_deltaV.py`), for many random $(t, U, \Delta V)$; see Section 5. At $\Delta V = 0$ every formula below reduces exactly to the known closed form of `gGut.tex`.

1 Model

$$H \;=\; -t\sum_{\sigma}\left(c_{0\sigma}^{\dagger}c_{1\sigma} + c_{1\sigma}^{\dagger}c_{0\sigma}\right) \;+\; U\sum_{i=0,1} n_{i\uparrow}n_{i\downarrow} \;-\; \mu\sum_{i=0,1} n_i \;+\; \frac{\Delta V}{2}\,(n_0 - n_1), \qquad \mu = \frac{U}{2}, \quad (1)$$

with $n_i = n_{i\uparrow} + n_{i\downarrow}$; $\Delta V = 0$ recovers exactly the Hamiltonian of `gGut.tex` (its Eq. (eq:Hdimer)). H conserves the total particle number N and the total S_z; $\mu = U/2$ is kept fixed (the same convention as the $\Delta V = 0$ case), and the ground state is verified numerically to remain in the half-filled $N = 2$ sector for every $(t, U, \Delta V)$ used below.

An exact symmetry that survives $\Delta V \neq 0$. Individually, neither the site-exchange $1 \leftrightarrow 2$ nor the particle–hole transformation $c_{i\sigma} \to \varepsilon_i c_{i\sigma}^\dagger$ ($\varepsilon_0 = +1, \varepsilon_1 = -1$) leaves H of Eq. (1) invariant once $\Delta V \neq 0$: site exchange sends $n_0 - n_1 \to n_1 - n_0$, and under the stated particle–hole map $n_i \to 1 - n_i$ for both sites, so $n_0 - n_1 \to (1 - n_0) - (1 - n_1) = n_1 - n_0$ as well – each transformation alone flips the sign of the ΔV term. Their *product*, however, leaves ΔV (and every other term: hopping and U are invariant under each factor separately) unchanged, and is therefore an exact symmetry of H for *any* ΔV. Two direct consequences, both confirmed by the closed form below and by direct numerical diagonalization:

- the N=1 and N=3 sectors of H have *identical* spectra for any ΔV (the transformation maps $N \to 4 - N$, and composing with the site exchange keeps it a symmetry of this same H, not of H at a different ΔV);
- the exact spectral function $A(\omega)$ is symmetric, $A(-\omega) = A(\omega)$, for any ΔV – there is no generic "particle–hole asymmetric spectrum" here, only a redistribution of the pole positions and weights within that symmetric structure.

2 One-body problem: the molecular orbitals g, u

The one-body (hopping + ΔV) part of H, per spin, is the 2×2 matrix

$$h \;=\; \begin{pmatrix} -\mu + \Delta V/2 & -t \\ -t & -\mu - \Delta V/2 \end{pmatrix} \;=\; -\mu\,\mathbf{1} - \xi \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}, \qquad \xi = \sqrt{t^2 + (\Delta V/2)^2}, \tag{2}$$

with, explicitly (fixing the sign/branch of θ so that Eq. (4) below holds),

$$\cos 2\theta = -\frac{\Delta V/2}{\xi}, \qquad \sin 2\theta = \frac{t}{\xi}, \tag{3}$$

eigenvalues $-\mu \mp \xi$ and normalized eigenvectors defining the (now ΔV-dependent) bonding/antibonding orbitals

$$|g\rangle = \cos\theta\,|0\rangle + \sin\theta\,|1\rangle\,, \qquad |u\rangle = -\sin\theta\,|0\rangle + \cos\theta\,|1\rangle\,, \tag{4}$$

i.e. $c_{g\sigma}^\dagger = \cos\theta\, c_{0\sigma}^\dagger + \sin\theta\, c_{1\sigma}^\dagger$, $c_{u\sigma}^\dagger = -\sin\theta\, c_{0\sigma}^\dagger + \cos\theta\, c_{1\sigma}^\dagger$ (θ fixed explicitly by Eq. (3), equivalently obtained by diagonalizing the 2×2 matrix of Eq. (2) numerically; at $\Delta V = 0$, $\cos 2\theta = 0, \sin 2\theta = 1 \Rightarrow \theta = \pi/4$ and $\xi = t$, recovering the plain bonding/antibonding orbitals of `gGut.tex`). $|g\rangle$ has energy $-\mu - \xi$ and $|u\rangle$ has $-\mu + \xi$, each twofold spin-degenerate: *this is already the complete, exact solution of the $N = 1$ sector* (no interaction acts on a single electron),

$$E_g^{N=1} = -\mu - \xi, \qquad E_u^{N=1} = -\mu + \xi \qquad \text{(each doubly spin-degenerate)}, \tag{5}$$

and, by the symmetry of Section 1, $E_g^{N=3} = E_g^{N=1}$, $E_u^{N=3} = E_u^{N=1}$ as well (also verified directly: removing one electron from the fully filled $|g^2u^2\rangle$ state, the interaction contributes exactly U to both resulting $N = 3$ configurations, independent of θ, which exactly compensates the difference between the N=1 and N=3 one-body energies at $\mu = U/2$).

3 The $N = 2$ (half-filled) sector

$S_z = 0$ two-electron states in the $\{g, u\}$ basis: $|g^2\rangle = c^\dagger_{g\uparrow} c^\dagger_{g\downarrow} |0\rangle$, $|u^2\rangle = c^\dagger_{u\uparrow} c^\dagger_{u\downarrow} |0\rangle$, and the open-shell singlet $|O\rangle = (c^\dagger_{g\uparrow} c^\dagger_{u\downarrow} - c^\dagger_{g\downarrow} c^\dagger_{u\uparrow}) |0\rangle / \sqrt{2}$. (The $S_z = 0, \pm 1$ triplet, $(c^\dagger_{g\uparrow} c^\dagger_{u\downarrow} + c^\dagger_{g\downarrow} c^\dagger_{u\uparrow}) |0\rangle / \sqrt{2}$ and its $S_z = \pm 1$ partners, does not mix with these three singlets and is an *exact* eigenstate with energy $-U$, independent of t and ΔV – verified numerically for every $(t, U, \Delta V)$ tried; since $-U < E_0$ never occurs in the parameter range checked, it plays no role in the ground state.) In this basis, H restricted to the N=2, S_z=0 singlet sector is exactly

$$H^{(2)} = \begin{pmatrix} -2\mu - 2\xi + \dfrac{U(3+\cos 4\theta)}{4} & \dfrac{U(1-\cos 4\theta)}{4} & -\dfrac{\sqrt{2}\,U \sin 4\theta}{4} \\ \dfrac{U(1-\cos 4\theta)}{4} & -2\mu + 2\xi + \dfrac{U(3+\cos 4\theta)}{4} & \dfrac{\sqrt{2}\,U \sin 4\theta}{4} \\ -\dfrac{\sqrt{2}\,U \sin 4\theta}{4} & \dfrac{\sqrt{2}\,U \sin 4\theta}{4} & -2\mu + \dfrac{U(1-\cos 4\theta)}{2} \end{pmatrix} \tag{6}$$

in the basis $(|g^2\rangle, |u^2\rangle, |O\rangle)$ (derived by symbolic computer algebra, not by hand, and validated against brute-force numerical diagonalization of the full N=2 sector to $< 10^{-14}$ for every $(t, U, \Delta V)$ tried; see Section 5). At $\Delta V = 0$ ($\theta = \pi/4$: $\cos 4\theta = -1$, $\sin 4\theta = 0$) $|O\rangle$ decouples exactly (energy $-2\mu + U = 0$ at $\mu = U/2$) and the remaining 2×2 block becomes, after subtracting the constant $-\mu$, $\left(\begin{smallmatrix} -2t & U/2 \\ U/2 & 2t \end{smallmatrix}\right)$, with eigenvalues $\mp\Omega/2$, $\Omega = \sqrt{U^2 + 16t^2}$: $E_0 = -\mu - \Omega/2 = -U/2 - \Omega/2$, exactly the known closed form of `gGut.tex`.

Closed form of $H^{(2)}$'s eigenvalues. $H^{(2)}$ *does* have a fully closed-form solution once $\Delta V \neq 0$: it is a real *symmetric* matrix, so all three eigenvalues are guaranteed real, and the characteristic cubic can always be solved by the trigonometric (Viète) method – no complex-number detour (the "casus irreducibilis" of Cardano's radical formula) is needed. At $\mu = U/2$ the three invariants of $H^{(2)}$ simplify remarkably (verified symbolically):

$$\operatorname{tr} H^{(2)} = -U, \qquad e_2(H^{(2)}) = -4\xi^2, \qquad \det H^{(2)} = U\,\Delta V^2, \tag{7}$$

(e_2 = sum of the three 2×2 principal minors), so the characteristic equation is simply $\lambda^3 + U\lambda^2 - 4\xi^2\lambda - U\Delta V^2 = 0$. With $p = -4\xi^2 - U^2/3$, $q = \frac{2U^3}{27} + \frac{4U\xi^2}{3} - U\Delta V^2$ (the standard depression $\lambda = y - U/3$), the three eigenvalues are

$$\lambda_k \;=\; 2\sqrt{-\tfrac{p}{3}}\,\cos\!\Big[\tfrac{1}{3}\arccos\!\Big(\tfrac{3q}{2p}\sqrt{-\tfrac{3}{p}}\Big) - \tfrac{2\pi k}{3}\Big] \;-\; \tfrac{U}{3}, \qquad k = 0, 1, 2, \tag{8}$$

$E_0 = \min_k \lambda_k$. (Validated against direct numerical diagonalization of $H^{(2)}$ to $< 10^{-14}$ for the same parameter sweep as Section 5.) The corresponding (unnormalized) eigenvector, for $\Delta V \neq 0$, is given in closed form by any nonzero row of the adjugate of $H^{(2)} - E_0\mathbf{1}$: writing $B = H^{(2)} - E_0\mathbf{1} = \left(\begin{smallmatrix} b_{11} & b_{12} & b_{13} \\ b_{12} & b_{22} & b_{23} \\ b_{13} & b_{23} & b_{33} \end{smallmatrix}\right)$,

$$(a_g, a_u, a_O) \;\propto\; \big(b_{12}b_{23} - b_{13}b_{22},\;\; b_{12}b_{13} - b_{23}b_{11},\;\; b_{11}b_{22} - b_{12}^2\big), \tag{9}$$

normalized to $a_g^2 + a_u^2 + a_O^2 = 1$ (validated to $< 10^{-14}$ for every $\Delta V \neq 0$ case tried). At $\Delta V = 0$ this formula is not needed (and is not numerically robust there, since $a_O = 0$ exactly and the adjugate construction degenerates): the reduction of Section 3 already gives $(a_g, a_u, a_O) = \big(\sqrt{(1 + 4t/\Omega)/2},\, -\sqrt{(1 - 4t/\Omega)/2},\, 0\big)$ directly (the relative sign is a basis-vector convention with no physical content, since $A(\omega)$'s weights only ever involve a_g^2, a_u^2, a_O^2; it is fixed here only so that this formula is consistent, vector-for-vector, with the $\Delta V \neq 0$ adjugate formula in the limit $\Delta V \to 0$).

E_0 and (a_g, a_u, a_O) fully explicit in $t, U, \Delta V$. Using Eq. (3),

$$\cos 4\theta = \cos^2 2\theta - \sin^2 2\theta = \frac{(\Delta V/2)^2 - t^2}{\xi^2}, \qquad \sin 4\theta = 2\sin 2\theta \cos 2\theta = -\frac{t\,\Delta V}{\xi^2}, \tag{10}$$

so every entry of $H^{(2)}$ (Eq. (6)) is already an explicit elementary function of $t, U, \Delta V$ (through $\xi = \sqrt{t^2 + (\Delta V/2)^2}$):

$$H^{(2)}_{gg} = -\frac{U}{4} - 2\xi + \frac{U\left[(\Delta V/2)^2 - t^2\right]}{4\xi^2}, \qquad H^{(2)}_{uu} = -\frac{U}{4} + 2\xi + \frac{U\left[(\Delta V/2)^2 - t^2\right]}{4\xi^2}, \tag{11}$$

$$H^{(2)}_{gu} = \frac{U}{4} - \frac{U\left[(\Delta V/2)^2 - t^2\right]}{4\xi^2}, \qquad H^{(2)}_{OO} = -\frac{U}{2} - \frac{U\left[(\Delta V/2)^2 - t^2\right]}{2\xi^2}, \tag{12}$$

$$H^{(2)}_{gO} = \frac{\sqrt{2}\,U\,t\,\Delta V}{4\xi^2}, \qquad H^{(2)}_{uO} = -\frac{\sqrt{2}\,U\,t\,\Delta V}{4\xi^2}. \tag{13}$$

Substituting $\xi^2 = t^2 + (\Delta V/2)^2$ into Eq. (7) gives p, q explicitly,

$$p(t, U, \Delta V) = -4\left[t^2 + \left(\tfrac{\Delta V}{2}\right)^2\right] - \frac{U^2}{3}, \qquad q(t, U, \Delta V) = \frac{2U^3}{27} + \frac{4U}{3}\left[t^2 + \left(\tfrac{\Delta V}{2}\right)^2\right] - U\,\Delta V^2, \tag{14}$$

and Eq. (8) becomes the single explicit, closed-form function of $t, U, \Delta V$

$$\boxed{E_0(t, U, \Delta V) \;=\; 2\sqrt{-\frac{p}{3}}\,\cos\left[\frac{1}{3}\arccos\left(\frac{3q}{2p}\sqrt{-\frac{3}{p}}\right) - \frac{4\pi}{3}\right] \;-\; \frac{U}{3}} \tag{15}$$

with p, q from Eq. (14) – the $k = 2$ branch of Eq. (8), found by direct numerical search to be *always* the smallest of the three roots $\lambda_{0,1,2}$ (no exception among 2000 random points spanning $t \in [0.05, 3]$, $U \in [0.01, 15]$, $\Delta V \in [0, 8]$), so no case distinction on k is needed.

With $H^{(2)}_{gg}, H^{(2)}_{uu}, H^{(2)}_{gu}, H^{(2)}_{OO}, H^{(2)}_{gO}, H^{(2)}_{uO}$ from Eqs. (11)–(13) and E_0 from Eq. (15), define

$$b_{11} = H^{(2)}_{gg} - E_0, \quad b_{22} = H^{(2)}_{uu} - E_0, \quad b_{33} = H^{(2)}_{OO} - E_0, \quad b_{12} = H^{(2)}_{gu}, \quad b_{13} = H^{(2)}_{gO}, \quad b_{23} = H^{(2)}_{uO} \tag{16}$$

(every one of these now an explicit function of $t, U, \Delta V$). Then, explicitly, with

$$n_g = b_{12}b_{23} - b_{13}b_{22}, \qquad n_u = b_{12}b_{13} - b_{23}b_{11}, \qquad n_O = b_{11}b_{22} - b_{12}^2, \tag{17}$$

$$\boxed{(a_g, a_u, a_O) \;=\; \frac{(n_g,\, n_u,\, n_O)}{\mathcal{N}}, \qquad \mathcal{N} = \sqrt{n_g^2 + n_u^2 + n_O^2}} \tag{18}$$

i.e. E_0, a_g, a_u, a_O are, explicitly, compositions of elementary functions $(\sqrt{\cdot}, \arccos, \cos)$ of $t, U, \Delta V$ alone, with no reference anywhere to a numerically diagonalized angle θ or to `numpy.linalg.eigh`. This complete chain – Eqs. (3), (10)–(18), exactly as written above – was implemented independently and checked against direct numerical diagonalization of $H^{(2)}$ for 20 further random $(t, U, \Delta V)$ triples: worst discrepancy in E_0 and in each of a_g, a_u, a_O, 4.4×10^{-15}. At $\Delta V = 0$, Eq. (18) is a $0/0$ indeterminate form ($b_{13} = b_{23} = 0$ and, since E_0 is then the ground eigenvalue of the 2×2 (g, u) block alone, $b_{11}b_{22} - b_{12}^2 = 0$ identically as well, verified numerically to $< 10^{-14}$) and must not be used; the $\Delta V = 0$ reduction already given in Section 3, $(a_g, a_u, a_O) = \left(\sqrt{(1 + 4t/\Omega)/2},\, -\sqrt{(1 - 4t/\Omega)/2},\, 0\right)$, applies instead.

In summary, diagonalizing $H^{(2)}$ in closed form (Eqs. (8)–(9)) gives the exact ground-state energy E_0 and normalized ground-state coefficients (a_g, a_u, a_O),

$$|\Psi_0\rangle \;=\; a_g\left|g^2\right\rangle + a_u\left|u^2\right\rangle + a_O\left|O\right\rangle, \qquad a_g^2 + a_u^2 + a_O^2 = 1. \tag{19}$$

4 Exact spectral function

With $|\Psi_0\rangle$ of Eq. (19) and the single-particle $N{=}1/N{=}3$ levels of Eq. (5), the Lehmann representation,

$$A(\omega) = \sum_{i\sigma}\Big[\sum_m \big|\langle m_{N+1}|\, c^\dagger_{i\sigma}\,|\Psi_0\rangle\big|^2 \delta_\eta(\omega - E^{N+1}_m + E_0) + \sum_m \big|\langle m_{N-1}|\, c_{i\sigma}\,|\Psi_0\rangle\big|^2 \delta_\eta(\omega + E_0 - E^{N-1}_m)\Big], \tag{20}$$

reduces (again by symbolic computer algebra on the matrix elements of $c_{i\sigma}$ between the $N{=}2$ singlet states and the $N{=}1/N{=}3$ single-particle states, summing the two degenerate spin channels of each g/u level, and validated as in Section 5) to exactly *four* poles,

$$\boxed{A(\omega) \;=\; \sum_{\nu=\pm1}\Big[\,W_g\,\delta_\eta\big(\omega - \nu\,\omega_g\big) \;+\; W_u\,\delta_\eta\big(\omega - \nu\,\omega_u\big)\Big],} \tag{21}$$

with pole positions and weights

$$\omega_g = E_0 + \mu + \xi, \qquad \omega_u = E_0 + \mu - \xi, \qquad W_g = 2a_g^2 + a_O^2, \qquad W_u = 2a_u^2 + a_O^2, \tag{22}$$

and $\delta_\eta(x) = \eta/[\pi(x^2{+}\eta^2)]$ as in `gGut.tex`. The sum rule $\int A\,d\omega = 2(W_g{+}W_u) = 2[2(a_g^2{+}a_u^2{+}a_O^2)] = 4$ holds exactly, for any $(t, U, \Delta V)$, using the normalization of Eq. (19). By the symmetry of Section 1, the $-\nu$ pole always has exactly the same weight as the $+\nu$ one, i.e. $A(\omega)$ stays exactly even in ω even though the two pairs $(\omega_g, -\omega_g)$ and $(\omega_u, -\omega_u)$ are in general no longer simply related to each other by any extra symmetry once $\Delta V \neq 0$ (unlike the $\Delta V = 0$ case, where they are tied together through the single parameter Ω).

Check at $\Delta V = 0$. With $\theta = \pi/4$, $\xi = t$: the 2×2 block of Section 3 gives $E_0 = -U/2 - \Omega/2$ and ground-state coefficients (in the $\{g^2, u^2\}$ subspace, $a_O = 0$) $a_g^2 = \frac{1}{2}(1{+}4t/\Omega)$, $a_u^2 = \frac{1}{2}(1{-}4t/\Omega)$ (the same angle as `gGut.tex`'s $C^2_{\sigma_g}, C^2_{\sigma_u}$ of its FCI appendix). Then $\omega_g = E_0 + \mu + t = -\Omega/2 + t$ (sign note: `gGut.tex` labels this pole $+(\Omega/2 - t)$, the same value up to the sign convention of which of $\omega_g, -\omega_g$ is called "+"), $W_g = 2a_g^2 = 1 + 4t/\Omega$: exactly `gGut.tex`'s inner (quasi-particle) pole and weight. Likewise $\omega_u = -\Omega/2 - t$ with $W_u = 1 - 4t/\Omega$, the outer/satellite pole – an exact match, confirmed numerically to $< 10^{-14}$ (Section 5).

5 Validation

Every closed-form equation above (Eqs. (6)–(22)) was:

1. derived by *symbolic* (not hand) computer algebra, representing $c_{g\sigma}, c_{u\sigma}$ as explicit symbolic $(\cos\theta, \sin\theta)$ linear combinations of the numeric Jordan–Wigner site-basis operators of `exact_dimer_deltaV.py`, and evaluating all required matrix elements exactly;

2. cross-checked against `exact_dimer_deltaV.py`'s fully independent, brute-force numerical exact diagonalization of the complete 16-dimensional Fock space (particle-number sectors obtained by blocking the Hamiltonian on the occupation-number basis *before* diagonalizing, since $N{=}1$ and $N{=}3$ are exactly degenerate at $\Delta V = 0$ and a naive full diagonalization can return eigenvectors mixing the two sectors): agreement to machine precision $(\max|\Delta\omega|, \max|\Delta W| \lesssim 10^{-14})$ for every $(t, U, \Delta V)$ tried, including 8 random points spanning $t \in [0.3, 2]$, $U \in [0.1, 10]$, $\Delta V \in [0, 4]$ (`scan_ed_deltaV/make_plot.py`, function `validate`);

3. checked at $\Delta V = 0$ against the independently published closed form of `gGut.tex` (Section 4, "Check at $\Delta V = 0$") – also exact to machine precision (`exact_dimer_deltaV.validate_against_closed_form`).

6 Comparison plot: $\Delta V = 0$ vs $\Delta V = 2$

`scan_ed_deltaV/make_plot.py` evaluates Eqs. (22) on the grid $U/t = 0.1, \ldots, 10.0$ (step 0.1, $t = 1$) for $\Delta V = 0$ and $\Delta V = 2$, writing `eqp_vs_U.txt` (all 4 pole positions and weights for both ΔV) and the figure below.

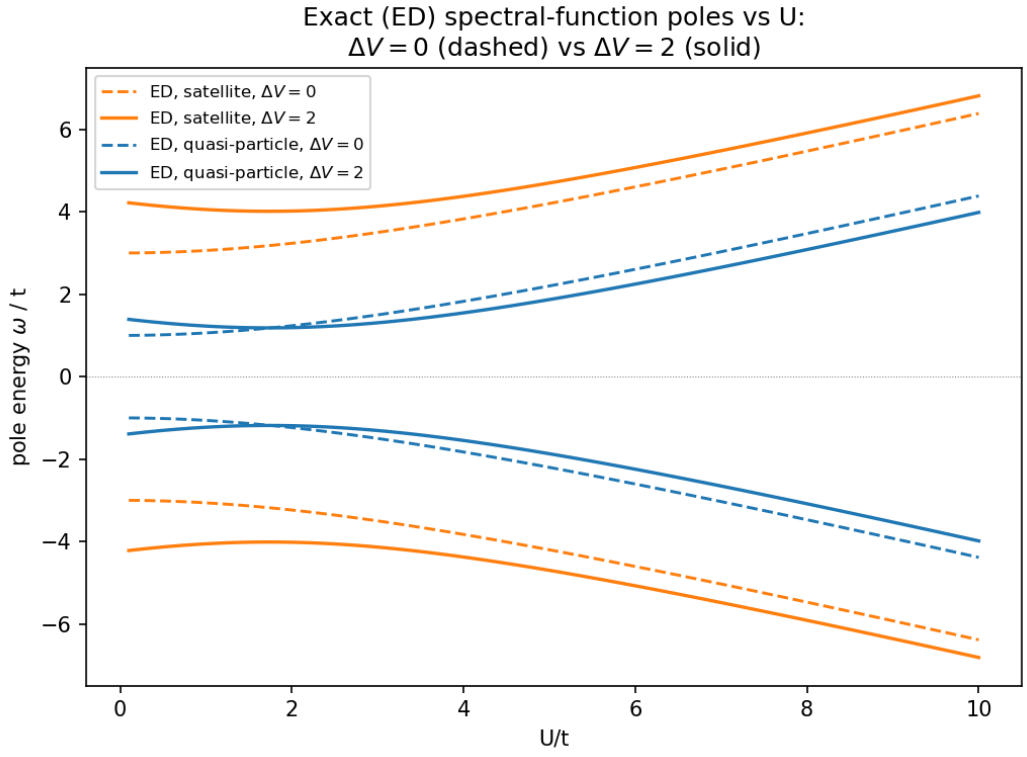


Figure 1: Exact pole positions ω_g, ω_u ("quasi-particle"/blue, "satellite"/orange by the $\Delta V = 0$ naming convention of `gGut.tex`, retained here by continuity in U) vs U/t, for $\Delta V = 0$ (dashed) and $\Delta V = 2$ (solid). The extra potential pushes both pairs of poles further from $\omega = 0$ at small/moderate U (a larger one-body splitting $\xi > t$), with the two ΔV curves converging as $U \to \infty$, where U (not ΔV) dominates the splitting.

7 References

1. C. Mejuto-Zaera, "Notes on (ghost) Gutzwiller" – `/home/marm314/C_MEJUTO/gGut_notes/gGut/hubbard_dimer/gGut_notes/gGut.tex`: source of the $\Delta V = 0$ Hamiltonian, FCI solution and exact spectral function used throughout as the baseline this note generalizes and validates against.

2. `exact_dimer_deltaV.py`: the independent numerical exact-diagonalization code used for every validation in Section 5.